# PREDICTIVE MAPPING OF *AEDES AEGYPTI* ECOLOGICALLY SUITABLE HABITATS: AN ENSEMBLE MODELLING APPROACH FOR CLIMATE-INFORMED VECTOR CONTROL

***Heury Ferr***
*Institute of Biological Sciences*
*Esperança ave., Samambaia Campus, Goiânia, Brasil, 74 690-900*
*heuryferr@gmail.com*

**Summary**

Aedes aegypti, the primary vector of arboviruses such as dengue, Zika, chikungunya, and yellow fever, remains a serious and persistent global public health concern, particularly in tropical and subtropical regions. This study aimed to overcome the challenge of identifying ecologically suitable habitats for A. aegypti by modeling its potential spatial distribution across a representative Brazilian region. To achieve this, four distinct species distribution modeling (SDM) techniques were applied: generalized additive models (GAM), generalized linear models (GLM), BIOCLIM, and maximum entropy (MaxEnt), in addition to an ensemble model that integrates their outputs to improve predictive performance. These models were calibrated using verified occurrence records and a comprehensive set of bioclimatic variables, allowing for ecologically meaningful and spatially explicit predictions. Among the individual techniques, GAM and MaxEnt provided the most accurate results, as determined by widely used evaluation metrics such as the area under the curve (AUC), the true skill statistic (TSS), and the correlation coefficient. Temperature seasonality (bio4) and precipitation of the wettest month (bio13) emerged as the most influential environmental predictors associated with the mosquito's spatial distribution. The ensemble model effectively reduced predictive uncertainty and improved reliability by combining the strengths of the individual approaches. These findings support the identification of high-risk areas for A. aegypti establishment and offer valuable guidance for designing targeted vector surveillance and control strategies, especially in the face of ongoing climate change.

**Keywords**: Aedes aegypti, species distribution, niche modelling, MaxEnt, suitability, vector surveillance.

**DOI: 10.21303/2504-5695.2025.003927**

## 1. Introduction

The mosquito *Aedes aegypti* is a primary vector for several significant viral diseases, including dengue, Zika, chikungunya, and yellow fever, posing a major global public health challenge [1]. These diseases result in substantial morbidity and mortality, particularly in tropical and subtropical regions [2]. The occurrence and population dynamics of *Aedes aegypti* are influenced by various environmental, climatic, and socioeconomic factors, such as temperature, precipitation, humidity, urbanization, and human mobility [3].

Understanding the spatial distribution of *Aedes aegypti* is crucial for developing effective vector control strategies and mitigating the impact of the diseases it transmits. Accurate predictions of its distribution can inform public health interventions, optimize resource allocation, and enhance preparedness for potential outbreaks [4]. Species distribution models (SDMs) are powerful tools that enable the prediction of species' potential distributions based on environmental variables. These models integrate ecological theory with statistical and machine learning techniques to provide insights into the habitat suitability and spatial patterns of species [5].

Utilizing distinct methods for species distribution modeling (SDM) is crucial for capturing the complex relationships between species and their environments. Different algorithms, such as MaxEnt, GLM, and GAM, each have unique strengths and limitations, which can lead to varying predictions when applied independently [6]. Combining multiple modeling approaches through ensemble methods can enhance prediction accuracy and robustness by integrating the strengths of individual models and reducing the biases inherent to any single method [7]. Ensemble models have been shown to provide more reliable and comprehensive insights into species distributions, improving the overall quality of ecological predictions and decision-making in conservation planning [8].

There are several metrics that can be used to compare the performance of different models. Metrics such as area under the receiver operating characteristic curve (AUC), true skill statistic (TSS), and correlation coefficient (COR) are widely used to evaluate the predictive performance of species distribution models (SDMs). AUC measures the model's ability to distinguish between presence and absence data, with values closer to 1 indicating excellent discrimination capacity [9]. TSS assesses the balance between sensitivity and specificity, offering a robust metric for model evaluation in ecological studies [10]. Similarly, COR provides insight into the linear relationship between observed and predicted values, while metrics like the F1-score and Matthews correlation coefficient (MCC) quantify the trade-offs between precision and recall, making them particularly useful in imbalanced datasets [11]. These metrics collectively enable a comprehensive assessment of model accuracy and reliability in ecological applications.

This study aims to model the spatial distribution of *Aedes aegypti* incorporating multiple modeling approaches and a wide range of predictor variables, it is possible to generate robust predictions of Aedes aegypti distribution and identify key environmental determinants of its habitat suitability. These findings are expected to contribute to more targeted and effective vector control strategies, ultimately reducing the burden of Aedes-borne diseases.

## 2. Materials and methods

The occurrence data for *Aedes aegypti* were downloaded from the GBIF website, and the climatic variables were extracted from the worldclim.org website, with 30 seconds of resolution. Initially, it is possible to clean the dataset by removing duplicate occurrence records and points with missing latitude or longitude values. This resulted in 18,314 occurrence records worldwide.

Next, using the set of predictor variables, it is possible to conduct a multicollinearity analysis, specifically employing the Variance Inflation Factor (VIF) to detect and manage collinearity in environmental predictors. This approach is crucial in SDM to ensure more reliable model predictions by excluding predictor variables with high VIF values [12, 13]. Nine of twenty variables showed high VIF values (VIF $\geq$ 10) and were excluded, leaving 11 environmental variables as predictors.

Different modeling approaches were employed to study the distribution of *Aedes aegypti*, including maximum entropy (MaxEnt), BIOCLIM, generalized linear models (GLM), and generalized additive models (GAM). MaxEnt is widely used due to its robustness and ability to handle presence-only data [14]. BIOCLIM uses climatic envelopes to define suitable habitats but may lack flexibility compared to advanced techniques [15]. GLMs and GAMs model the relationship between species presence and environmental variables, with GAMs providing greater flexibility through non-linear smooth functions [16, 17].

The ensemble model approach was also used, combining the aforementioned approaches for greater robustness [7]. Distribution models were built using 30% of the dataset as a test set, while the remaining data was used for training. 1000 background points were used to configure pseudo-absences. The analyses were performed in R version 4.3.3, utilizing the dismo [18], rgbif [19], raster [20], sdm [21], tidyverse [22], and writexl [23] packages. Maps were generated using QGIS Desktop version 3.34.4.

## 3. Results and discussion

The **Table 1** presents the evaluation metrics used to assess the predictive performance of the models, including AUC (Area Under the ROC Curve), COR (Correlation Coefficient), Deviance, TSS (True Skill Statistic), MCC (Matthews Correlation Coefficient), and F1-score. These metrics provide a comprehensive assessment of model accuracy, discrimination capacity, and reliability in predicting the spatial distribution of *Aedes aegypti*.

The performance metrics for species distribution models (SDMs) were evaluated for four algorithms: generalized additive models (GAM), generalized linear models (GLM), BIOCLIM, and maximum entropy (MaxEnt) (**Table 1**). GAM demonstrated the highest AUC (0.941) and TSS (0.786), indicating superior predictive performance among the models. MaxEnt also performed well, with an AUC of 0.931 and a TSS of 0.771. In contrast, GLM and BIOCLIM had lower AUC values (0.904 and 0.914, respectively), with BIOCLIM presenting the lowest correla-

tion (COR = 0.199) and the highest deviance (5.006), reflecting its limitations in capturing the species' distribution. Across all models, GAM achieved the highest F1-score (0.979), further corroborating its robustness in predictive accuracy (**Table 1**).

The comparative performance of habitat suitability models (GAM, GLM, Bioclim, and Maxent) reveals significant variability in predicted maximum suitability values and spatial distributions for Aedes aegypti. Generalized Additive Models (GAM) and Generalized Linear Models (GLM) predict the highest habitat suitability scores, reaching a maximum of 1, suggesting extensive favorable conditions. Conversely, Bioclim predicts a maximum suitability of 0.77, while Maxent's predictions peak at 0.65, reflecting more conservative estimations of habitat suitability (**Fig. 1**).

The results from the correlation and AUC metrics indicate that different environmental variables exhibit varying levels of importance in modeling species distribution (**Fig. 2**). Based on the correlation metric, bio9 (21.0), bio8 (18.3), bio13 (17.7), and bio3 (14.8) showed the highest values, suggesting greater relevance in the model. Conversely, according to the AUC metric, bio13 (9.8) emerged as the most important variable, followed by bio9 (4.1), bio5 (3.4), bio8 (3.4), and bio19 (2.9). Other variables, such as bio2, bio14, and elevation, exhibited lower relative contributions across both metrics.

Spatially, GAM and GLM delineate Brazil, the southern United States, most of the African continent, and southern Eurasia as regions with high suitability. However, GAM identifies some areas of lower suitability in northern Africa and parts of North America, highlighting the model's nuanced predictions. In contrast, Bioclim exhibits the most restrictive spatial distribution, identifying limited regions in Brazil, Africa, and North America as suitable habitats. Maxent predictions are broader than those of Bioclim, with suitability values above 0.6 extending across much of South America, Africa, southern Eurasia, and a significant portion of Australia (**Fig. 1**). Thus, let's also generate the forecast by combining these multiple modeling approaches through ensemble model (**Fig. 3**).

**Table 1**

Performance metrics for species distribution models

| Model | AUC | COR | Deviance | TSS | MCC | F1 |
|---|---|---|---|---|---|---|
| GAM | 0.941 | 0.839 | 0.139 | 0.786 | 0.682 | 0.979 |
| GLM | 0.904 | 0.785 | 0.183 | 0.699 | 0.554 | 0.967 |
| BIOLCLIM | 0.914 | 0.199 | 5.006 | 0.741 | 0.556 | 0.964 |
| MAXENT | 0.931 | 0.801 | 0.904 | 0.771 | 0.650 | 0.976 |

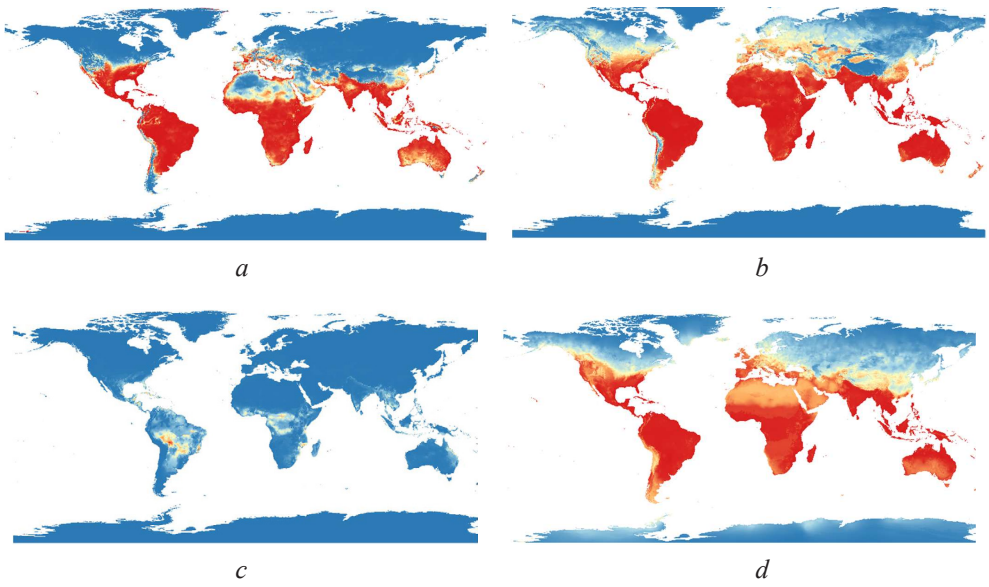



**Fig. 1.** Predicted habitat suitability for Aedes aegypti generated by four modeling algorithms: *a* – GAM – generalized additive model; *b* – GLM – generalized linear model; *c* – BIOCLIM; *d* – MaxEnt

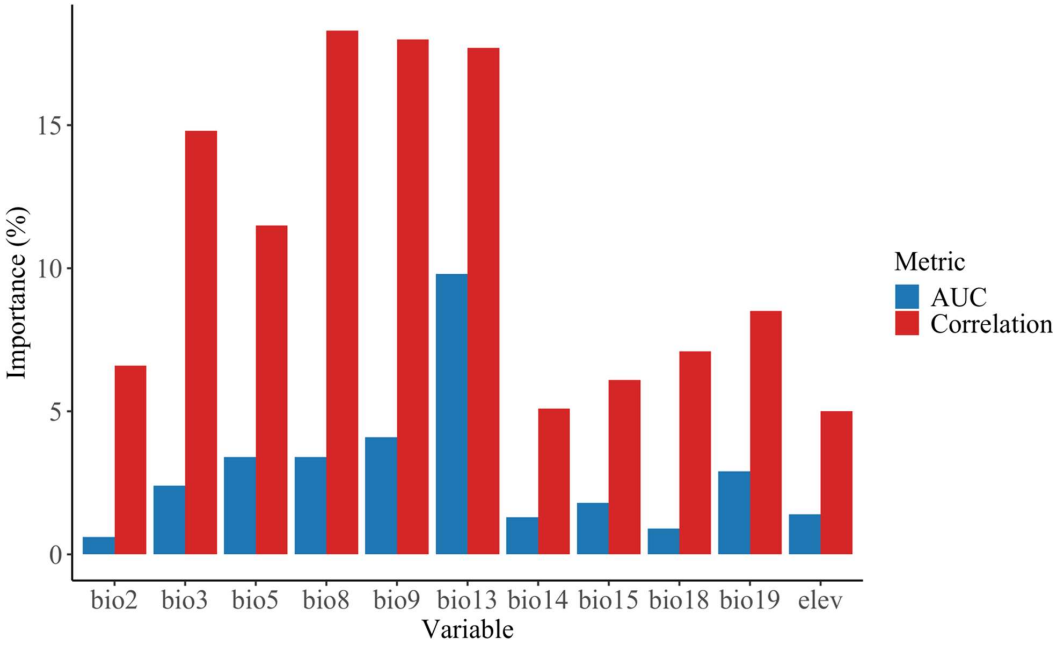



**Fig. 2.** Variable importance based on correlation and AUC metrics: red – correlation with model outputs; blue – contribution to AUC

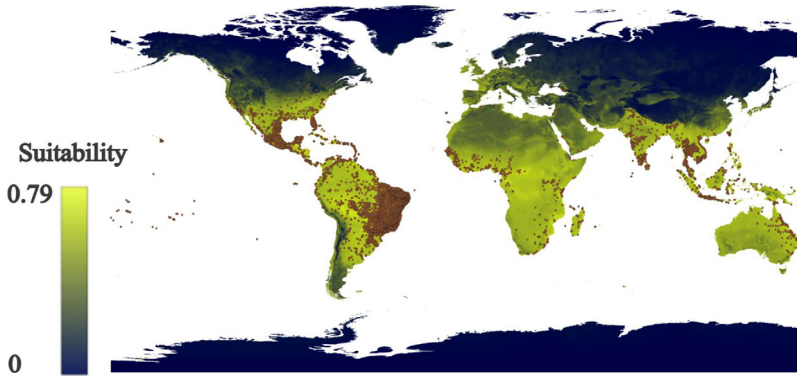



**Fig. 3.** Predicted habitat suitability for Aedes aegypti based on ensemble modeling

The variability observed across modeling algorithms underscores not only the ecological complexity of predicting *Aedes aegypti* distribution but also the critical public health implications of these predictions. Generalized additive models (GAM) and MaxEnt demonstrated the most robust performance, with GAM's non-linear smoothing capabilities allowing it to capture intricate ecological interactions. Similarly, MaxEnt, widely recognized for its ability to handle presence-only data and diverse ecological conditions, highlighted regions of concern with broader suitability predictions. These tools are particularly valuable in the context of *Aedes aegypti*, as its adaptability to diverse environmental conditions poses challenges for control and prevention strategies [1, 6].

In contrast, GLM and Bioclim, while effective in specific scenarios, showed limitations when applied to a species with a broad ecological niche such as *Aedes aegypti*. GLM's linear framework may oversimplify the species-environment relationships, and Bioclim's envelope-based predictions restrict its capacity to capture subtle environmental gradients. For example, Bioclim's conservative projections of suitability excluded regions where mosquito proliferation has already been observed, potentially underestimating risk areas. This could have

significant implications for vector control programs, particularly in emerging disease hotspots where proactive surveillance is crucial [20].

These discrepancies among modeling approaches align with expectations based on their methodological differences. GAM and GLM are parametric methods that capture complex interactions between predictors, often resulting in higher suitability scores. Bioclim, relying on environmental envelopes, typically yields more conservative results by emphasizing climatic similarity to known presences. MaxEnt's machine-learning approach allows it to delineate suitability based on presence-only data but may generate broader predictions due to its ability to incorporate marginal environments into its predictions [6, 14].

To reconcile these differences and minimize the biases inherent to individual models, an ensemble model was constructed. Ensemble approaches integrate predictions from multiple algorithms to produce robust and consensus-based suitability maps [7]. This integration is particularly important for public health applications, as it reduces uncertainty and enhances the reliability of spatial predictions, providing a more comprehensive understanding of suitable habitats for *Aedes aegypti*. The ensemble approach employed in this study integrates the strengths of individual models, producing a more robust and comprehensive prediction. This consensus-based method is particularly relevant for health-focused applications, where the cost of underestimating habitat suitability can result in delayed responses to outbreaks. Additionally, the ensemble approach allows for a nuanced understanding of uncertainty, enabling decision-makers to target areas where predictions converge and allocate resources effectively [7]. Moving forward, incorporating real-time climatic data and mosquito surveillance into ensemble frameworks could further refine these predictions, enhancing their utility in dynamic and rapidly changing environments.

From a medical perspective, the discrepancies in predictions are not merely academic. *Aedes aegypti* is a primary vector for several arboviruses, including dengue, Zika, chikungunya, and yellow fever. Understanding the spatial dynamics of its habitat suitability is essential for prioritizing vector control measures and optimizing the allocation of public health resources. The findings emphasize the importance of accurate, high-resolution suitability maps in guiding interventions, such as insecticide deployment, community education, and vaccine distribution, especially in regions identified as emerging or high-risk zones by predictive models [24].

The correlation and AUC metrics revealed distinct patterns of variable importance, reflecting differences in how each metric evaluates the relevance of variables in the model. Bio9 (temperature seasonality) was highly correlated, indicating its strong direct influence on species distribution, while bio13 (precipitation of the wettest month) exhibited the highest predictive power in the AUC metric, suggesting a more specific relationship with habitat suitability. These findings suggest that variables related to climatic variation and seasonality are critical for understanding the species' ecology. Bio13, which stood out in both metrics, may serve as a key variable, and its ecological interpretation should consider water availability during critical periods. The divergence between metrics highlights the importance of employing complementary approaches when prioritizing variables in species distribution modeling studies.

Ultimately, this study demonstrates that methodological choices have direct consequences for public health planning. Predictive tools must balance ecological precision with practical applicability, ensuring that the generated insights are actionable for disease prevention and control. As arboviral diseases continue to expand their geographic range due to globalization and climate change, the integration of ecological and medical perspectives will be indispensable in safeguarding global health.

## 4. Conclusions

This study underscores the critical role of selecting and integrating modeling algorithms to predict habitat suitability for *Aedes aegypti*, a vector with profound public health implications. The superior performance of GAM and MaxEnt highlights their utility in capturing the ecological complexity of species distribution, while the limitations of GLM and Bioclim emphasize the need for caution when applying simpler models to dynamic and heterogeneous environments. By

adopting an ensemble approach, it was demonstrated how combining predictions from multiple models enhances reliability and reduces uncertainty, producing actionable insights for vector control and disease prevention. The results indicate that bio13 and bio9 are key variables for understanding the species distribution, highlighting the importance of precipitation and climate seasonality. The findings have immediate relevance for public health strategies, particularly in guiding interventions to mitigate the spread of arboviral diseases such as dengue, Zika, and chikungunya. Moving forward, the integration of real-time climatic data and on-ground mosquito surveillance into predictive frameworks will be crucial in adapting to the evolving challenges posed by globalization and climate change, ensuring that proactive measures can effectively safeguard vulnerable populations.

**Conflict of interest**

The authors declare that they have no conflict of interest in relation to this research, whether financial, personal, authorship or otherwise, that could affect the research and its results presented in this paper.

**Financing**

The research was performed without financial support.

**Data availability**

Data will be made available on reasonable request.

**Use of artificial intelligence**

The authors confirm that they did not use artificial intelligence technologies when creating the current work.

**References**

[1] Dengue and severe dengue. WHO. Available at: http://www.who.int/news-room/fact-sheets/detail/dengue-and-severe-dengue

[2] Bhatt, S., Gething, P. W., Brady, O. J., Messina, J. P., Farlow, A. W., Moyes, C. L. et al. (2013). The global distribution and burden of dengue. Nature, 496 (7446), 504–507. https://doi.org/10.1038/nature12060

[3] Kraemer, M. U. G., Sinka, M. E., Duda, K. A., Mylne, A., Shearer, F. M., Brady, O. J. et al. (2015). The global compendium of Aedes aegypti and Ae. albopictus occurrence. Scientific Data, 2 (1). https://doi.org/10.1038/sdata.2015.35

[4] Messina, J. P., Brady, O. J., Golding, N., Kraemer, M. U. G., Wint, G. R. W., Ray, S. E. et al. (2019). The current and future global distribution and population at risk of dengue. Nature Microbiology, 4 (9), 1508–1515. https://doi.org/10.1038/s41564-019-0476-8

[5] Elith, J., Leathwick, J. R. (2009). Species Distribution Models: Ecological Explanation and Prediction Across Space and Time. Annual Review of Ecology, Evolution, and Systematics, 40 (1), 677–697. https://doi.org/10.1146/annurev.ecolsys.110308.120159

[6] Elith, J., H. Graham, C., P. Anderson, R., Dudík, M., Ferrier, S., Guisan, A. et al. (2006). Novel methods improve prediction of species' distributions from occurrence data. Ecography, 29 (2), 129–151. Portico. https://doi.org/10.1111/j.2006.0906-7590.04596.x

[7] Araujo, M., New, M. (2007). Ensemble forecasting of species distributions. Trends in Ecology & Evolution, 22 (1), 42–47. https://doi.org/10.1016/j.tree.2006.09.010

[8] Marmion, M., Parviainen, M., Luoto, M., Heikkinen, R. K., Thuiller, W. (2008). Evaluation of consensus methods in predictive species distribution modelling. Diversity and Distributions, 15 (1), 59–69. https://doi.org/10.1111/j.1472-4642.2008.00491.x

[9] Swets, J. A. (1988). Measuring the Accuracy of Diagnostic Systems. Science, 240 (4857), 1285–1293. https://doi.org/10.1126/science.3287615

[10] Allouche, O., Tsoar, A., Kadmon, R. (2006). Assessing the accuracy of species distribution models: prevalence, kappa and the true skill statistic (TSS). Journal of Applied Ecology, 43 (6), 1223–1232. https://doi.org/10.1111/j.1365-2664.2006.01214.x

[11] Powers, D. M. (2011). Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation. International Journal of Machine Learning Technology, 2 (1), 37–63. https://doi.org/10.48550/arXiv.2010.16061

[12] Guisan, A., Thuiller, W. (2005). Predicting species distribution: offering more than simple habitat models. Ecology Letters, 8 (9), 993–1009. https://doi.org/10.1111/j.1461-0248.2005.00792.x

[13] Dormann, C. F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G. et al. (2012). Collinearity: a review of methods to deal with it and a simulation study evaluating their performance. Ecography, 36 (1), 27–46. https://doi.org/10.1111/j.1600-0587.2012.07348.x

[14] Phillips, S. J., Anderson, R. P., Schapire, R. E. (2006). Maximum entropy modeling of species geographic distributions. Ecological Modelling, 190 (3-4), 231–259. https://doi.org/10.1016/j.ecolmodel.2005.03.026

[15] Nix, H. (1986). A biogeographic analysis of Australian elapid snakes. Atlas of Elapid Snakes of Australia. Canberra: Australian Flora and Fauna Series, 4–15. Available at: https://ia801408.us.archive.org/25/items/atlaselapidsnak00busb/atlaselapidsnak00busb_bw.pdf

[16] Guisan, A., Edwards, T. C., Hastie, T. (2002). Generalized linear and generalized additive models in studies of species distributions: setting the scene. Ecological Modelling, 157 (2-3), 89–100. https://doi.org/10.1016/s0304-3800(02)00204-1

[17] Hastie, T. J., Tibshirani, R. J. (2017). Generalized Additive Models. Routledge. https://doi.org/10.1201/9780203753781

[18] Hijmans, R. J., Phillips, S., Leathwick, J., Elith, J. (2010). dismo: Species Distribution Modeling [dataset]. In CRAN: Contributed Packages. The R Foundation. https://doi.org/10.32614/cran.package.dismo

[19] Chamberlain, S., Oldoni, D., Waller, J. (2012). rgbif: Interface to the Global Biodiversity Information Facility API [dataset]. In CRAN: Contributed Packages. The R Foundation. https://doi.org/10.32614/cran.package.rgbif

[20] Hijmans, R. J. (2010). raster: Geographic Data Analysis and Modeling [dataset]. In CRAN: Contributed Packages. The R Foundation. https://doi.org/10.32614/cran.package.raster

[21] Naimi, B., Araújo, M. B. (2016). sdm: a reproducible and extensible R platform for species distribution modelling. Ecography, 39 (4), 368–375. https://doi.org/10.1111/ecog.01881

[22] Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R. et al. (2019). Welcome to the Tidyverse. Journal of Open Source Software, 4 (43), 1686. https://doi.org/10.21105/joss.01686

[23] Ooms, J. (2017). writexl: Export Data Frames to Excel "xlsx" Format [dataset]. In CRAN: Contributed Packages. The R Foundation. https://doi.org/10.32614/cran.package.writexl

[24] Hemingway, J., Field, L., Vontas, J. (2002). An Overview of Insecticide Resistance. Science, 298 (5591), 96–97. https://doi.org/10.1126/science.1078052

*Received 15.05.2025*
*Received it revised form 11.07.2025*
*Accepted 03.08.2025*
*Published 14.08.2025*

*© The Author(s) 2025*
*This is an open access article*
*under the Creative Commons CC BY license*
*https://creativecommons.org/licenses/by/4.0/*

***How to cite:*** *Ferr, H. (2025). Predictive mapping of Aedes aegypti ecologically suitable habitats: an ensemble modelling approach for climate-informed vector control. EUREKA: Life Sciences, 3, 3–9. https://doi.org/10.21303/2504-5695.2025.003927*